



NEKO PHYSICAL INTELLIGENCE

MECHANICAL & ELECTROMECHANICAL MANUFACTURING MASTER

Controlled architecture baseline for geometry, structure, manufacturing, integration, mass properties, safety, FAT and production release

DOCUMENT STATUS

Revision R00 consolidates the engineering architecture frozen through M232-M295. It is the master baseline for the next CAD/data closure phase. Fields dependent on real supplier CAD, measured mass, verified load cases or certified component data remain explicitly TBD - they are not invented.

FIELD	VALUE
Document ID	NEKO-MFG-MASTER
Revision	R00
Date	2026-09-29
Owner	ADNEKO · NEKO Physical Intelligence
Document class	Controlled engineering / manufacturing
Release level	Architecture baseline; CAD/data closure pending
Procurement authority	NOT AUTHORIZED until applicable gates pass

CORE RULE

Open → resolve → verify → evidence → close → next. No estimated value becomes a production specification without traceable evidence.

1. Document Control and Master Scope

This document is the single controlled baseline for the mechanical and electromechanical architecture of NEKO Physical Intelligence. It consolidates the manufacturing rules already closed in engineering and defines the evidence required before prototype procurement, build, FAT and production release.

REV	DATE	SCOPE	STATUS
R00	2026-09-29	Initial consolidation of M232-M295	CONTROLLED BASELINE

1.1 Authority hierarchy

- NEKO-ASM-MASTER is the sole master CAD authority once released.
- Released drawings govern manufactured dimensions and tolerances.
- Approved BOM governs exact purchased components and alternatives.
- As-built record governs the actual configuration of each physical unit.
- This master governs design/release logic and required verification evidence.

1.2 Release philosophy

GATE	MEANING
DF-0	Concept architecture
DF-1	Interfaces frozen
DF-2	Geometry frozen
DF-3	Mass / dynamics validated
DF-4	Manufacturing package frozen
DF-5	Prototype build authorized
DF-6	FAT passed
DF-7	Production release

CURRENT CONTROL POINT

Architecture requirements are consolidated. Exact supplier geometry, mass properties, final J02 brake selection, counterbalance geometry and other data-dependent parameters must now be closed before DF-3 / procurement release.

1.3 Canonical status vocabulary

STATE	USE
WIP	Active engineering work
REVIEW	Ready for technical verification
FROZEN	Technical content fixed; evidence complete for scope
RELEASED	Authorized for downstream use
OBSOLETE	Superseded; not permitted for new work

2. Geometry Baseline and Rigid-Body Decomposition

2.1 Canonical kinematic baseline

REFERENCE	VALUE
J01	0 mm reference
J02	110 mm
J03	280 mm
J04	530 mm
TCP	650 mm
J01 → J02	110 mm
J02 → J03	170 mm
J03 → J04	250 mm
J04 → TCP	120 mm
J01 → TCP nominal	650 mm

J01 and J02 are defined as non-intersecting axes. The canonical right-handed frame chain is $\{B\} \rightarrow \{J01\} \rightarrow \{J02\} \rightarrow \{J03\} \rightarrow \{J04\} \rightarrow \{TCP\}$.

2.2 Rigid-body ownership

RB	MANUFACTURABLE BODY
RB-001	Shoulder/base moving body associated with J01
RB-002	Shoulder-roll carrier / proximal moving assembly
RB-003	Upper-arm structural assembly
RB-004	Elbow moving assembly
RB-005	Forearm structural assembly
RB-006	Wrist + gripper carrier assembly
RB-007	Gripper / tool assembly

MASS OWNERSHIP RULE

Every physical mass belongs to exactly one rigid body or declared fixed subsystem. No item may be counted twice or remain outside the mass reconciliation.

2.3 Joint stack reference

JOINT	ACTUATOR	NOMINAL / PEAK	MASS	RATIO
J01	AKH70-48 V1.0	74 / 222 N·m	1.396 kg	48:1
J02	AKH70-16 V1.0	26 / 78 N·m	0.879 kg	16:1
J03	AK60-39 V3.0	24 / 72 N·m	0.750 kg	39:1
J04	AK45-10 V3.0	2.5 / 7 N·m	0.262 kg	10:1

PROCUREMENT STATUS

The joint stack is a geometry/reference baseline only. Purchase remains blocked until the required fit/load/interface evidence is complete.

3. Datum System, CAD Contract and Geometry Acceptance

3.1 Joint datum architecture

DATUM	FUNCTION
A	Housing / primary structural interface
B	Primary bearing seat / concentric structural reference
C	True kinematic rotation axis
D	Reducer / actuator interface plane
E	Output encoder reference plane
F	Output flange / next rigid-body interface

A/B/C govern structure and concentricity; D/E govern drivetrain and sensing; F governs continuation of the kinematic chain. The reducer is not assumed to substitute for a dedicated load-bearing architecture unless its capability is explicitly verified.

3.2 Mandatory CAD mass-property record

FIELD	REQUIRED CONTENT
Identity	CAD filename, revision, extraction timestamp, CAD tool/version
Kinematics	parent_joint, child_joint, joint_axis, zero-angle transform, positive rotation, mechanical range
Mass	mass_kg
COM	[x,y,z] m in declared frame
Inertia	Ixx, Iyy, Izz, Ixy, Ixz, Iyz [kg·m ²]
Traceability	property_frame, verified_by

FRAME INTEGRITY

A mass, COM or inertia tensor without an explicit reference frame is invalid for the dynamics model.

3.3 Geometry acceptance - G3 RELEASED FOR MANUFACTURING

CHECK	PASS CONDITION
Geometry	Real closed geometry; no placeholder production interfaces
Datums	Declared and traceable
Joint axes	Explicit reference geometry
Interfaces	Real mounting planes / bolt patterns / registers
Purchased CAD	Official or controlled supplier geometry; no scaling
Bearings / fasteners	Real interfaces and installation strategy
Routing / stops	Cable passages, hard stops and service clearances represented
Material / density	Assigned for mass-property extraction
Revision	Controlled revision; no ambiguous FINAL/LATEST naming
DFM/DFA	Manufacturable, assemblable and inspectable

3.4 Prohibited release conditions

- VERIFY BEFORE DRILLING on frozen interfaces.
- Uncontrolled fit-as-required corrections.
- Supplier CAD scaled away from 1.000.
- Hidden mass corrections outside the master model.
- Undefined zero-angle or joint-axis convention.
- Uninspectable critical tolerances.

4. Mass Properties, J02 Closure and Counterbalance

4.1 Exact mass reconciliation

The moving assembly must satisfy $\Sigma \text{mass}(\text{RB-001} \dots \text{RB-007}) = \text{modeled moving mass}$, including actuators, bearings, brackets, structure, covers, fasteners, encoders, brakes, moving cable, connectors, sensors, gripper and defined payload ownership. Estimated mass is not permitted in critical downstream components at final dynamics release.

4.2 J02 critical transverse COM

Final J02 brake selection is driven by the perpendicular distance from the J02 axis to the real downstream center of mass. The required quantity is $r_{\text{COM},\text{J02}}$ = perpendicular distance(J02 axis → downstream COM), extracted from the real assembly CAD and transformed into {J02}.

J02 BRAKE

5 N·m remains a CANDIDATE ONLY. It must not be released as the final brake rating until gravity, dynamic stopping, power-loss behavior, margin and thermal/duty constraints are resolved from real mass properties.

4.3 J02 governing load cases

CASE	DESCRIPTION
LC-J02-01	Static gravity hold
LC-J02-02	Maximum gravitational moment
LC-J02-03	Nominal motion acceleration
LC-J02-04	Controlled deceleration
LC-J02-05	Emergency stop
LC-J02-06	Power loss + brake engagement
LC-J02-07	Maximum defined payload
LC-J02-08	Counterbalance degraded/unavailable, if applicable

4.4 Counterbalance closure

The prior compensation model remains an engineering reference, but final implementation requires manufacturable anchor geometry, lever arm, spring/gas element selection, stroke, preload, travel envelope and interference verification. The mechanism must avoid over-center singularity, negative assistance, excessive bearing side-load and service obstruction.

5. Structural Load Path and Load Cases

5.1 Primary load path

Payload / Tool → RB-007 → RB-006 → RB-005 → RB-004 → RB-003 → RB-002 → RB-001 → ARM_TO_TORSO_INTERFACE → TORSO PRIMARY FRAME → MAIN CHASSIS.

STRUCTURAL RULE

Motors, encoders, PCBs, connectors, cable harnesses, cosmetic covers and sensor housings are not primary load-bearing elements

unless a dedicated analysis explicitly releases them for that function.

5.2 Joint load categories

CATEGORY	MEANING
T	Torsion
M	Bending moment
Fr	Radial force
Fa	Axial force
Shock	Transient / impact load

5.3 Structural load cases

CASE	DESCRIPTION
LC-S01	Static self weight
LC-S02	Nominal payload
LC-S03	Maximum defined payload
LC-S04	Worst gravity configuration
LC-S05	Maximum acceleration
LC-S06	Controlled stop
LC-S07	Emergency stop
LC-S08	Power loss
LC-S09	Hard-stop event
LC-S10	Service / transport handling

5.4 Stiffness and modal behavior

Structural sign-off requires both strength and stiffness. Excessive deflection can produce TCP error, vibration, control oscillation, perception misalignment and collision risk even when stress remains below yield. Modal behavior must be checked so operational control/motion frequencies do not excite flexible structural modes.

6. Material, Process and Tolerance Architecture

6.1 Functional material families

CLASS	FUNCTION
S1	Primary structure
S2	Joint / bearing precision structure
S3	Secondary structure
S4	Electronics / internal support
S5	External body / cosmetic
S6	Wear / contact component
S7	Thermal component
S8	Elastomer / damping

Material selection follows FUNCTION → LOAD → GEOMETRY → PROCESS → MATERIAL → FINISH. Exact alloy, grade, hardness and thickness are not frozen until the governing load and geometry are known.

6.2 Authorized manufacturing processes

PROCESS	PRIMARY USE / CONTROL
CNC machining	Precision bearing/joint interfaces, structural brackets, datums
Sheet metal	Electronics trays, panels, secondary supports; flat pattern required
Extrusion / profile	Chassis/frame elements with exact supplier section
Welding	Only where structural/economic advantage is clear; distortion controlled
Additive	Covers, guides, ducts, noncritical brackets; critical use requires explicit validation
Thermoform / molding	Candidate for production body shells after interfaces are stable

6.3 Tolerance classes

CLASS	EXAMPLES
T1 Precision	Bearing seats, shafts, encoder alignment, actuator pilot
T2 Structural	Structural interfaces, locating features, arm-to-torso interface
T3 Assembly	Covers, trays, brackets, cable supports
T4 Cosmetic	Shell gaps and visual surfaces

GD&T RULE

Critical geometry uses explicit position, perpendicularity, parallelism, flatness, runout/profile controls as appropriate. Notes such as “make aligned” are not release criteria.

7. Fasteners, Serviceability, Drawings and BOM

7.1 Fastener architecture

- Each structural joint declares whether fasteners act primarily in preload, tension, shear or combined loading.
- Precision location is not delegated to ordinary fasteners when a register/dowel/pilot is required.
- Service areas use inserts/captive hardware where repeated maintenance would damage base material.
- Every structural fastener receives a locking method and a torque source based on exact material, grade and thread condition.
- A single controlled NEKO TORQUE TABLE will be released after exact hardware is frozen.

7.2 Service levels

LEVEL	SCOPE
L1 FIELD	Covers, accessible sensors, external cables, compute/power modules
L2 TECHNICAL	Actuator, encoder, brake, bearing set, internal harness
L3 FACTORY	Precision structural joint rebuild, frame repair, critical alignment

7.3 Manufacturing drawing minimum content

MANDATORY FIELDS	DETAIL
Identity	NEKO part number, part name, revision, release status
Manufacturing	Material, process, finish
Geometry	Datums, critical dimensions, GD&T, threads/inserts
Control	Mass reference where critical; drawn/checked/released by

7.4 Mechanical BOM hierarchy

NEKO-01 → HEAD ASSEMBLY / NECK ASSEMBLY / TORSO ASSEMBLY / LEFT ARM / RIGHT ARM / CHASSIS / BODY PANELS / ELECTRONICS MOUNTING / FASTENER-HARDWARE KITS. Each joint module carries actuator, structural carrier, bearings, shaft/interface, encoder, brake if used, hard stop, fasteners, harness and covers.

7.5 Make / buy classification

CLASS	USE
MAKE	Manufactured by approved process
BUY	Exact purchased component
CONFIGURED BUY	Purchased item requiring controlled configuration
STANDARD HARDWARE	Standard fastener/hardware line

8. Electromechanical Integration and Harness Control

8.1 Functional domains

Each joint and subsystem separates POWER → ACTUATION → SENSING → SAFETY → COMMUNICATION. Bundles may be physically grouped only after electrical class, routing, shielding and service requirements are explicitly resolved.

8.2 Harness classes

CLASS	FUNCTION
PWR-HV / PWR-LV	Power distribution classes as applicable
MOTOR	Actuator power
BRAKE	Brake supply/control
ENCODER	Position feedback
SAFETY	Safety-relevant signals
DATA	Network / digital communications
SENSOR	Sensor I/O
GROUND / PE	Protective / bonding path
SERVICE	Dedicated service connections

8.3 Dynamic harness rules

- Respect minimum bend radius and flex/torsion life.
- Define neutral position, service loop and strain relief.
- Verify $\theta_{min} \rightarrow \theta_0 \rightarrow \theta_{max}$ without pinch, tension, rubbing or connector loading.
- Harness must follow joint motion; it must never become the mechanical motion limiter.

8.4 Connector control

Each connector declares mating pair, keying, retention, voltage/current class, signal class, shielding, service cycle, orientation and access. Mis-mating between incompatible adjacent functions must be prevented by physical keying or unmistakable controlled identification.

9. Grounding, EMC, Brake, Encoder and Safety Chain

9.1 Grounding / bonding

Protective earth, functional ground and signal reference are distinct concepts. Chassis bonding is controlled and cannot depend on paint, anodizing or accidental bearing contact. Where a moving joint does not provide reliable electrical continuity, a dedicated bonding path is required.

9.2 EMI / EMC physical segregation

- Motor switching, power conversion and brake switching are treated as primary noise sources.
- Encoders, cameras, audio, communication buses and precision sensors are protected victims.
- High-current forward/return paths are kept compact to minimize loop area.
- Power/motor routing is separated from encoder/sensor/data routing where practical.
- Filters and suppression devices must be located where they are electrically effective, not added as cosmetic end-stage patches.

9.3 Brake physical integration

A safety-relevant brake must define engaged/released logic, power-loss behavior, engagement/release delay, thermal/duty limits, holding capability, wear state and service strategy. Brake replacement must not destroy bearing alignment or encoder datum unless the service procedure explicitly includes recalibration.

9.4 Encoder / limits architecture

Motor-side position and output-side position are distinct where both exist. Each joint must define absolute/homing reference and separate CONTROL LIMIT, SAFETY LIMIT and MECHANICAL HARD STOP. The hard stop must prevent cable/encoder/structure destruction and is not a normal operating limit.

9.5 E-stop physical chain

E-STOP REQUEST → SAFETY LOGIC → MOTION ENERGY REMOVAL / SAFE TORQUE STATE → BRAKE ACTION
WHERE REQUIRED → CONTROLLED OR EMERGENCY DECELERATION → SAFE STATE. Electrical energy, stored DC bus energy, gravity, kinetic energy and counterbalance energy are treated separately. Reset never implies automatic motion resume.

10. Thermal, Packaging, Sensor, Audio and Power Architecture

10.1 Thermal zones

ZONE	CONTENT
TZ-01	Compute
TZ-02	Power
TZ-03	Joint actuation
TZ-04	Sensor / head
TZ-05	Battery / energy if applicable

Cooling follows conduct → spread → ventilate where possible. If forced airflow is required, it must have a real inlet → heat source → outlet path with serviceable filters/fans as applicable.

10.2 Internal packaging

- Heavy components stay near the structural center and as low as functionally practical.
- Power and compute domains are separated when it improves EMC, cooling, service and fault isolation.
- Frequent-service modules receive direct access; internal routing does not require structural disassembly.

- Cable trays, grommets, clips and strain relief define production routing; temporary adhesive routing is not a production solution.

10.3 Sensor integration

Sensors are classified as perception, proprioception, safety, environment or diagnostic. Critical cameras mount to a rigid repeatable sensor frame, not solely to the cosmetic face shell, so visual redesign does not invalidate calibration.

10.4 NEKO Voice physical audio

The microphone array and loudspeaker architecture must control acoustic path, structure-borne vibration, fan noise, actuator/gear noise and acoustic feedback. Head CAD will define exact locations; the electrical/physical architecture must preserve full-duplex conversation performance.

10.5 Power distribution

PRIMARY ENERGY SOURCE → MAIN PROTECTION → POWER DISTRIBUTION → SUBSYSTEM PROTECTION → LOAD. Every branch will declare voltage/current class, protection, connector, wire-sizing basis and isolation requirement after the electrical load budget is frozen.

11. System Mass, Global COM and Stability

11.1 Mass groups

ID	GROUP
MG-01	Head
MG-02	Neck
MG-03	Torso structure
MG-04	Left arm
MG-05	Right arm
MG-06	Compute
MG-07	Power
MG-08	Safety/control electronics
MG-09	Chassis
MG-10	Body panels
MG-11	Harness/connectors
MG-12	Payload/tooling
MG-13	Future mobility system if applicable

11.2 Source quality

SOURCE	RELEASE USE
MEASURED	Accepted
VENDOR-CERTIFIED	Accepted
CAD-DERIVED	Accepted with controlled material/density
CALCULATED	Accepted with traceable method
ESTIMATED	Provisional only; prohibited for final critical dynamics inputs

11.3 Configuration-dependent COM cases

CASE	CONFIGURATION
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CASE	CONFIGURATION
CG-01	Neutral
CG-02	Arms forward
CG-03	One arm extended
CG-04	Both arms extended
CG-05	Nominal payload
CG-06	Maximum defined payload
CG-07	Head extreme
CG-08	Power-off safe pose

11.4 Stability

Global COM is computed from the complete mass model in the master frame. Static stability evaluates the projected COM against the actual support polygon with explicit margin; dynamic stability adds inertial resultants from acceleration, deceleration, arm reaction and chassis motion. “Inside the polygon” alone is not a complete dynamic safety criterion.

12. Transport, Calibration, Environment and Maintenance

12.1 Transport / handling

- Operating and transport configurations are separate.
- Lifting points attach to the primary structural frame; never lift by arm, head, cover or sensor mount.
- Joint transport restraints may be required so shipping vibration is not carried solely by the operational brake/drivetrain.
- Shipping dimensions and restraint geometry remain TBD until final envelope is available.

12.2 Calibration chain

MECHANICAL ZERO → ENCODER REFERENCE → CALIBRATION OFFSET → SOFTWARE JOINT POSITION. Mechanical zero, encoder zero and software zero are not assumed identical. Post-service calibration is classified as NO CALIBRATION / LOCAL CALIBRATION / FULL KINEMATIC CALIBRATION.

12.3 Metrology classes

CLASS	METHOD CATEGORY
MET-1	Shop inspection
MET-2	Precision inspection / CMM-equivalent
MET-3	Assembly alignment fixtures / indicators
MET-4	System kinematic calibration

12.4 Environmental release

Operating temperature, humidity, dust/liquid exposure, vibration, shock and indoor/outdoor classification remain configuration-specific release data. No IP rating is claimed without design and validation evidence for that exact configuration.

12.5 Maintenance and spares

Inspection/replacement/lubrication/calibration intervals will come from final components and validation. Condition-based maintenance may use motor temperature, current trend, position deviation, brake behavior, vibration and error history. Spares are classified as SP-1 FIELD STOCK / SP-2 SERVICE CENTER / SP-3 FACTORY ONLY.

13. Traceability, FAT and Procurement Gate

13.1 Serialization / as-built

Each NEKO unit receives a serial identity and module revision traceability. The as-built record must preserve installed actuator, brake, encoder, electronics, hardware revision, firmware/software configuration where applicable, inspection results and authorized substitutions.

13.2 Mechanical FAT

FAT	CHECK
FAT-MECH-01	Visual / workmanship
02	Dimensions
03	Fasteners / torque evidence
04	Joint free motion
05	Mechanical limits
06	Cable sweep
07	Structural integrity
08	Payload condition
09	Brake behavior
10	Thermal observation
11	Calibration
12	Safety chain

13.3 Electromechanical FAT

- Continuity and polarity.
- Grounding / bonding.
- Connector identification and harness routing.
- Encoder feedback and sensor channels.
- Brake command and power-loss behavior.
- E-stop chain and controlled recovery.
- Every test records PASS / FAIL plus evidence.

13.4 Procurement release rule

DO NOT BUY BEFORE RELEASE

A line is procurement-authorized only when exact P/N, manufacturer, quantity, technical fit, mechanical fit, electrical fit, load fit, CAD fit and approved revision are all controlled. Long-lead pressure does not override an unresolved critical interface.

CLASS	MEANING
A	Critical long lead
B	Important
C	Standard
D	Commodity

14. M296 - Manufacturing Master Consolidation

14.1 Controlled document set

DOCUMENT / DATASET	ROLE
NEKO-ASM-MASTER	Single authoritative master CAD assembly
NEKO-CAD-001...007	RB-001...RB-007 controlled manufactured geometry
NEKO-VND-J01...J04	Official/controlled supplier actuator geometry
NEKO-MECHANICAL-BOM	Hierarchical mechanical BOM
NEKO-TORQUE-TABLE	Controlled torque requirements after hardware freeze
NEKO-HARNESS-MASTER	Harness architecture and revisions
NEKO-CABLE-SCHEDULE	Cable identity/source/destination/function
NEKO-CONNECTOR-MATRIX	Mating/keying/electrical/service data
NEKO-POWER-DISTRIBUTION	Protected electrical distribution
NEKO-GROUNDING-BONDING	PE / chassis / functional grounding architecture
NEKO-JOINT-IO	Joint sensing/actuation/safety I/O
NEKO-SAFETY-CHAIN	Physical E-stop / inhibit / brake chain
NEKO-THERMAL-MAP	Heat sources, limits and cooling path
NEKO-INTERNAL-LAYOUT	Controlled internal packaging
NEKO-FAT-PACKAGE	Mechanical/electromechanical FAT procedures and evidence
NEKO-AS-BUILT	Per-unit installed configuration and calibration

14.2 Required file naming

Manufactured part example: NEKO-CAD-003-R00. Assemblies use controlled IDs such as NEKO-ASM-ARM and NEKO-ASM-MASTER. States are WIP / REVIEW / FROZEN / RELEASED / OBSOLETE. Ambiguous names such as FINAL_FINAL, LATEST or USE_THIS are prohibited.

14.3 Required next evidence package

PRIORITY	INPUT REQUIRED	PURPOSE
P0	Official STEP/X_T for AKH70-48 V1.0	J01 exact geometry / interface / mass reconciliation
P0	Official STEP/X_T for AKH70-16 V1.0	J02 geometry / COM / brake closure
P0	Official STEP/X_T for AK60-39 V3.0	J03 exact geometry
P0	Official STEP/X_T for AK45-10 V3.0	J04 exact geometry
P0	Current NEKO arm/master assembly if already available	Real geometry freeze and downstream mass properties
P1	Exact gripper/tool CAD + mass data	RB-007 / TCP / distal mass
P1	Brake candidate supplier data once shortlisted	J02 holding/dynamic/thermal validation
P1	Counterbalance component CAD/data once selected	Manufacturable geometry and load verification

THIS FILE

R00 is now the first controlled NEKO Manufacturing Master. Preserve it as the baseline. Future changes must increment revision and maintain traceability; do not overwrite the revision history.

15. Open Technical Inputs and Next Engineering Block

15.1 Data-dependent items intentionally OPEN

ITEM	WHY OPEN	CLOSURE EVIDENCE
Exact RB masses / COM / inertia	Requires final CAD/material/supplier data	CAD mass-property export + reconciliation
J02 final brake	Requires true transverse COM and stopping cases	Dynamics results + brake certified capability
Counterbalance geometry	Requires real axes, COM trajectory and selected element	CAD sweep + force/stroke/load validation
Exact materials / sections	Requires part loads and process closure	Analysis + released drawing
Bearing fits / detailed tolerances	Requires exact bearing P/N and load condition	Supplier data + tolerance stack
Wire gauges / currents	Requires final electrical load budget	Electrical calculation + harness release
Thermal loads / cooling	Requires actual component losses and packaging	Thermal analysis / test
Safety rating / category	Requires final use case, architecture and applicable standard	Formal safety analysis / verification
Environmental/IP claim	Requires exact enclosure and testing	Validated configuration evidence

15.2 Next block - M297 CAD Intake & Supplier Geometry Validation

The next engineering action is to ingest the four actuator supplier CAD models and/or the current NEKO master arm assembly, verify geometry integrity, map each model to the canonical datums, reconcile supplier mass, and populate the first real RB mass-property dataset. This is the bridge from architecture to numerical closure.

15.3 M297 acceptance sequence

STEP	ACTION
M297-01	Receive / identify source CAD and supplier revision
M297-02	Verify no scale conversion and units
M297-03	Map mounting planes, axis and interface geometry
M297-04	Record supplier mass vs CAD mass
M297-05	Integrate into NEKO-ASM-MASTER working assembly
M297-06	Extract RB mass/COM/inertia in declared frames
M297-07	Compute downstream COM relative to J02
M297-08	Run J02 governing torque/stop cases
M297-09	Close brake requirement
M297-10	Advance counterbalance geometry and Procurement Gate

CURRENT ACTION REQUIRED

No purchase is requested. The next external input, when available, is the official CAD/STEP/X_T package for the selected actuators or the current NEKO assembly CAD. All non-CAD-dependent manufacturing architecture is now consolidated in R00.